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INPUT SPECIFICATIONS

$Vi_min := 12\ V$

$Vi_max := 17\ V$

$Vi_nom := 13\ V$

OUTPUT SPECIFICATIONS

$Vo := 3.3\ V$

$Vrp := 50\ mV$

$Io_min := 0.5\ A$

$Io_max := 3\ A$

MIMIMUM AND MAXIMUM OUTPUT POWER

$Vp := 0.2\ V$

$Vd_fw := 0.6\ V$

$Po_min := (Vo + Vd_fw) \cdot Io_min = 1.95\ W$

$Po_max := (Vo + Vd_fw) \cdot Io_max = 11.7\ W$

$V0i := Vo + Vd_fw + Vp = 4.1\ V$

SWITCHING FREQUENCY

$fsw := 100\ kHz$

$Tsw := \frac{1}{fsw} = 1 \cdot 10^{-5}\ s$

MINIMUM AND MAXIMUM DUTY CYCLE

$Dmax := \frac{V0i}{Vi_min} = 0.3417$

$Ton_max := Tsw \cdot Dmax = 3.4167 \cdot 10^{-6}\ s$

$Dmin := \frac{V0i}{Vi_max} = 0.2412$

$Ton_min := Tsw \cdot Dmin = 2.4118 \cdot 10^{-6}\ s$

NOMINAL DUTY CYLE

$Dnom := \frac{V0i}{Vi_nom} = 0.3154$

$Ton_nom := Tsw \cdot Dnom = 3.1538 \cdot 10^{-6}\ s$

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CHOOSING THE DIODE (SCHOTTKY DIODE)

$Vdiode_max := Vi_max \cdot 1.15 = 19.55\ V$

$VF := 0.6\ V$

$rd := \frac{VF}{Io_max} = 0.2\ \Omega$

$Pdiode_max := Io_max \cdot VF \cdot (1 - Dmin) + rd \cdot Io_max^{2} \cdot (1 - Dmax) = 2.5509\ W$

INDUCTOR SIZING - continuous conduction mode

$Vf := \frac{V0i}{Ton_min} \cdot Tsw = 17\ V$

$Lo := \frac{(Vf - Vo) \cdot Ton_max}{2 \cdot Io_min} = 4.6808 \cdot 10^{-5}\ H$ inductance used: $L := 47 \cdot 10^{-6}\ H$

$\Delta Il := \frac{(Vf - Vo) \cdot Ton_min}{L} = 0.703\ A$

CAPACITOR SIZING

$C_min := \Delta Il \cdot \frac{1}{0.25 \cdot Vrp \cdot 8 \cdot fsw} = 7.03 \cdot 10^{-5}\ F$

$ESR_max := \frac{Vrp \cdot 0.75}{\Delta Il} = 0.0533\ \Omega$

MOSFET

$VDSS := Vi_max = 17\ V$

$Vdr := 9\ V$

$Idr := 3\ A$

$Rdr_on := \frac{Vdr}{Idr} = 3\ \Omega$

$Id_max := 3\ A$

FEEDBACK LOOP

$Vs := 3.3\ V$

$Co := 100 \cdot 10^{-6}\ F = 0.0001\ F$

$R01H := \frac{Vo}{Io_max} = 1.1\ \Omega$

$R01L := \frac{Vo}{Io_min} = 6.6\ \Omega$

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$$f_ph := \frac{1}{2 \cdot \pi \cdot \sqrt{L \cdot Co}} = 2321.5134 \text{ Hz}$$

$$f_z := \frac{1}{2 \cdot \pi \cdot ESR_max \cdot Co} = 29836.406 \text{ Hz}$$

$$Kpwr := \frac{Vi_max}{Vs} = 5.1515$$

$$Kpwrdbh := 20 \cdot \log_{10}(Kpwr) = 14.2387$$

$$fc := \frac{fsw}{5} = 20000 \text{ Hz}$$

$$Q := \frac{R01H}{2 \cdot \pi \cdot f_ph \cdot L} = 1.6045$$

$$H_pwrh(\omega) := Kpwr \cdot \frac{1 + \frac{i \cdot \omega}{2 \cdot \pi \cdot f_z}}{1 + \frac{i \cdot \omega}{2 \cdot \pi \cdot f_ph} \cdot \frac{1}{Q} + \left(\frac{i \cdot \omega}{2 \cdot \pi \cdot f_ph}\right)^2}$$

$$H_pwrhfc := \left|H_pwrh(2 \cdot \pi \cdot fc)\right| = 0.0845$$

$$\Phi_pwrfc := \frac{180}{\pi} \cdot \arg\left(H_pwrh(2 \cdot \pi \cdot fc)\right) = -141.9711$$

$$H_pwrhfcDB := 20 \cdot \log10\left(\left|H_pwrh(2 \cdot \pi \cdot fc)\right|\right) = -21.4654$$

$$R1 := 10 \text{ kohm}$$

$$M := 45$$

$$Boost := M - \Phi_pwrfc - 90 = 96.9711$$

$$HR := \frac{1}{H_pwrhfc} = 11.8378$$

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$$Kv := \left(\tan\left(\left(\frac{Boost \cdot \pi}{4 \cdot 180} \right) + \frac{45 \cdot \pi}{180} \right) \right)^2 = 6.9614$$

$$C2 := \frac{1}{2 \cdot \pi \cdot fc \cdot HR \cdot R1} = 6.7223 \cdot 10^{-11} \text{ F}$$

$$C1 := C2 \cdot (Kv - 1) = 4.0075 \cdot 10^{-10} \text{ F}$$

$$R2 := \frac{\sqrt{Kv}}{2 \cdot \pi \cdot fc \cdot C1} = 52392.7248 \ \Omega$$

$$R3 := \frac{R1}{Kv - 1} = 1677.4511 \ \Omega$$

$$C3 := \frac{1}{2 \cdot \pi \cdot fc \cdot \sqrt{Kv} \cdot R3} = 1.798 \cdot 10^{-9} \text{ F}$$

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